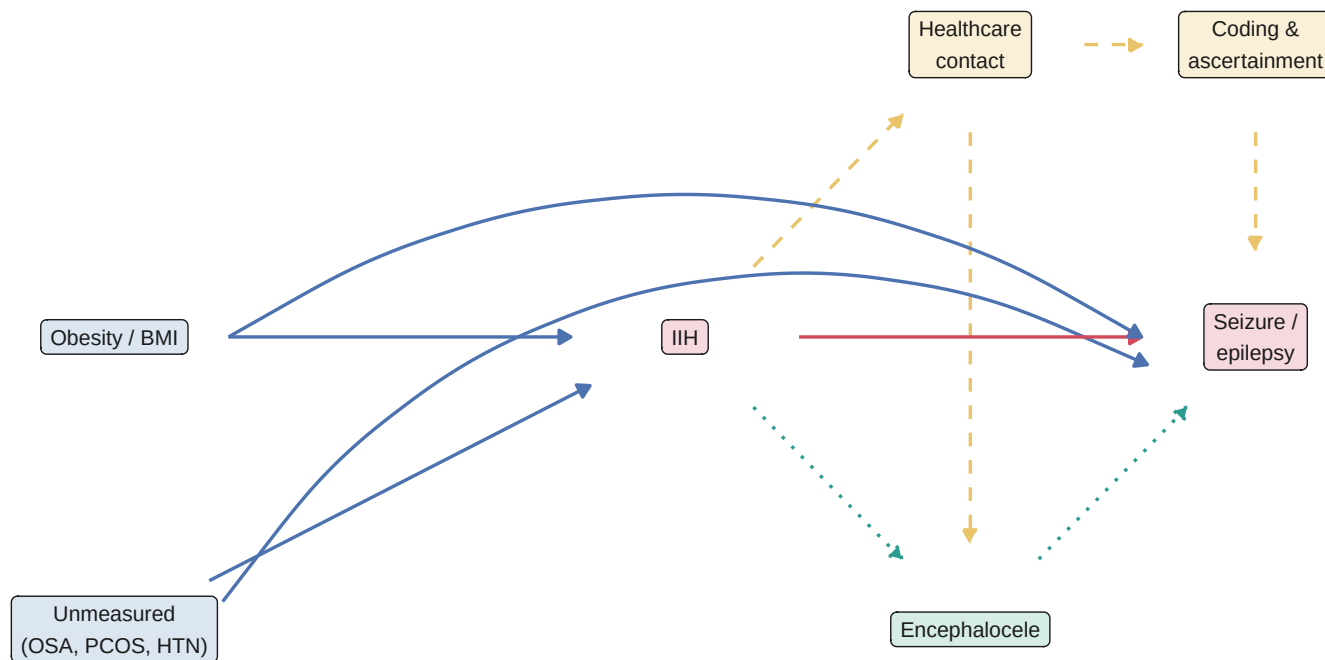


Assumed causal structure

Solid red = target effect. Dashed amber = the surveillance/ascertainment paths the design must break.



→ CAUSAL EFFECT OF INTEREST → bias → confounding → mediation

Matching closes the obesity/age/sex path. The encephalocele mediation path (dotted) is SUSPENDED: the mediator is measured in 2.5% of cases and 0% of controls. The amber paths are addressed by the negative-control outcome, not by adjustment -- conditioning on healthcare contact opens a collider path.